Software Engineering

LAB 7: Fuzzing for Input Validation Bugs

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BEFORE FIXING – Identification of Errors

1.Initial Test Execution – Test Collection Error

```
C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7>python -m pytest -v
========================= test session starts =========================
platform win32 -- Python 3.11.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7
collected 0 items / 1 error

============================== ERRORS ==============================
_____________ ERROR collecting test_loan_eligibility.py _____________
..\..\..\AppData\Roaming\Python\Python311\site-packages\_pytest\python.py:528: in importtestmodule
    mod = import_path(
..\..\..\AppData\Roaming\Python\Python311\site-packages\_pytest\pathlib.py:640: in import_path
    importlib.import_module(module_name)
C:\Program Files\Python311\Lib\importlib\__init__.py:126: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
<frozen importlib._bootstrap>:1206: in _gcd_import
    ???
<frozen importlib._bootstrap>:1178: in _find_and_load
    ???
<frozen importlib._bootstrap>:1149: in _find_and_load_unlocked
    ???
<frozen importlib._bootstrap>:690: in _load_unlocked
    ???
..\..\..\AppData\Roaming\Python\Python311\site-packages\_pytest\assertion\rewrite.py:176: in exec_module
    source_hash, co = _rewrite_test(fn, self.config)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
..\..\..\AppData\Roaming\Python\Python311\site-packages\_pytest\assertion\rewrite.py:339: in _rewrite_test
    tree = ast.parse(source, filename=strfn)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Program Files\Python311\Lib\ast.py:50: in parse
    return compile(source, filename, mode, flags,
E     File "C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 7
E       @given(st.text() | st.none())
E       ^
E   IndentationError: expected an indented block after function definition on line 5
======================= short test summary info =======================
ERROR test_loan_eligibility.py
!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!
========================== 1 error in 0.85s ===========================
```

2.Identification of Indentation Error in Test File

```
C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7>python -m pytest -v
========================= test session starts =========================
platform win32 -- Python 3.11.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7
collected 0 items / 1 error

============================== ERRORS ==============================
_____________ ERROR collecting test_loan_eligibility.py _____________
ImportError while importing test module 'C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py'.
Hint: make sure your test modules/packages have valid Python names.
Traceback:
C:\Program Files\Python311\Lib\importlib\__init__.py:126: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
test_loan_eligibility.py:1: in <module>
    from hypothesis import given, strategies as st
E   ModuleNotFoundError: No module named 'hypothesis'
====================== short test summary info ======================
ERROR test_loan_eligibility.py
!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!
========================= 1 error in 0.25s ==========================
```

3.Missing Hypothesis Library – Import Error

```
C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7>python -m pytest -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.11.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7
plugins: hypothesis-6.170.15
collecting 0 items                                                  collected 3 items

test_loan_eligibility.py::test_parse_income_no_crash FAILED [ 33%]
test_loan_eligibility.py::test_monthly_emi_no_crash FAILED [ 66%]
test_loan_eligibility.py::test_pan_category_no_crash FAILED [100%]

========================= FAILURES =========================
_____________ test_parse_income_no_crash _____________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 6, in test_parse_income_no_crash
  |     def test_parse_income_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7\test_loan_eligibility.py", line 8, in test_parse_income_no_crash
    |     parse_income(s)
    |   File "C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7\loan_eligibility.py", line 4, in parse_income
    |     return int(income.replace(',', ''))
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
```

4.Initial Fuzz Testing Results – 3 Tests Failed

```
    +------------------------------------
================== short test summary info ==================
FAILED test_loan_eligibility.py::test_parse_income_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 su...
FAILED test_loan_eligibility.py::test_monthly_emi_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 su...
FAILED test_loan_eligibility.py::test_pan_category_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 su...
=================== 3 failed in 12.82s ====================
```

Hypothesis_Test_Cases

File Edit View

```
from hypothesis import given, strategies as st
from loan_eligibility import parse_income, monthly_emi, pan_category


@given(st.text() | st.none())
def test_parse_income_no_crash(s):
    try:
        parse_income(s)
    except Exception as e:
        assert False, f"parse_income crashed with: {e}"


@given(st.text() | st.none())
def test_monthly_emi_no_crash(s):
    try:
        monthly_emi(s)
    except Exception as e:
        assert False, f"monthly_emi crashed with: {e}"


@given(st.text() | st.none())
def test_pan_category_no_crash(s):
    try:
        pan_category(s)
    except Exception as e:
        assert False, f"pan_category crashed with: {e}"
```

2. AFTER FIXING – Successful Execution

5.Final Fuzz Testing Results – All 3 Tests Passed

```
C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7>python -m pytest -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.11.2, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_loan_eligibility.py::test_parse_income_no_crash PASSED                                                                          [ 33%]
test_loan_eligibility.py::test_monthly_emi_no_crash PASSED                                                                           [ 66%]
test_loan_eligibility.py::test_pan_category_no_crash PASSED                                                                          [100%]

============================================================ 3 passed in 4.24s =============================================================
```

6.Manual Execution of Corrected Loan Eligibility Application

```
C:\Users\PRIYA\Downloads\L6_L7_26_loan_eligibility\Lab 7>python loan_eligibility.py
==== Running parse_income ====
Enter monthly income (e.g. 50,000): 75000
Income: 75000

==== Running monthly_emi ====
Enter principal,rate,years (e.g. 500000,10,5): 800000,8,4
Monthly EMI: 19530.34

==== Running pan_category ====
Enter a PAN number (e.g. ABCPE1234F): ABCPE1234F
PAN Category: Individual
```